

QUANTUM MECHANICS

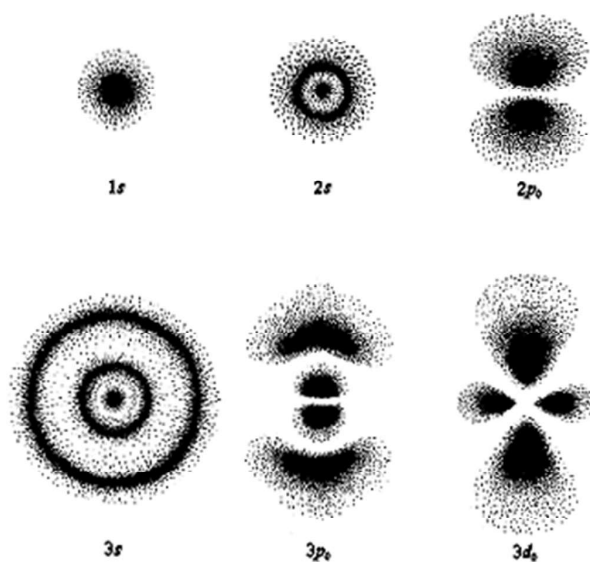
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End – Term Report

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1 Vector Spaces, Operators and Dirac Notation

Linear algebra is the language of quantum mechanics. The state vector Ψ that describes a quantum system is thought of as a vector in an infinite-dimensional vector space. Observables act as linear operators. Hence, in this section, we develop the required mathematical formalism needed to study quantum mechanics.

1.1 Vector Spaces

Vector Space is a set of **vectors** and **scalars** (numbers) that scale the vectors.

The scalars may either belong to the real numbers $\mathbb{R}$ or the complex numbers $\mathbb{C}$. We will use $\mathbb{F}$ to denote either $\mathbb{R}$ or $\mathbb{C}$.

Vectors are abstract quantities that satisfy the following properties. For vector space V ,

- They have an addition (+) operation such that $u + v \in V$ for each $u, v \in V$ (Closure under addition).
- They have a scalar multiplication operation such that $av \in V$ for every $a \in \mathbb{F}$, $v \in V$ (Closure under scalar multiplication).
- $u + v = v + u \in V$, for every $u, v \in V$ (Commutativity of addition).
- $u + (v + w) = (u + v) + w$ and $(ab)u = a(bu)$ for any $u, v, w \in V$, $a, b \in \mathbb{F}$ (Associativity).
- There is a vector $0 \in V$ such that $0 + u = u$ for all $u \in V$ (existence of additive identity).
- For each $u \in V$, there is a $v \in V$ such that $u + v = 0$ (existence of additive inverse).
- There exists scalar 1 such that $1v = v$ for every $v \in V$ (existence of multiplicative identity).
- $a(u + v) = au + av$ and $(a + b)v = av + bv$ for all $a, b \in \mathbb{F}$ and $u, v \in V$ (Distributivity).

Note that vectors are treated as abstract entities. Any object that satisfies the above properties is called a vector. Hence, it does not make sense to say whether a vector is 'real' or 'complex'. However, a vector space is said to be a **real vector space** or **complex vector space** based on whether the scalars in that space belong to the real numbers or the complex numbers.

Subspace of a vector space V is defined as a subset of V that is itself a vector space.

Span of a set of vectors $(v_1, v_2, \dots, v_n)$ is defined as the set of all linear combinations of these vectors, i.e.

$$\text{span}(v_1, v_2, \dots, v_n) = \{a_1v_1 + a_2v_2 + \dots + a_nv_n : a_i \in \mathbb{F}\}$$

Intuitively, span is the set all vectors that can be "reached" by the given set of vectors.

A set of vectors is said to be **linearly independent** if $a_1v_1 + a_2v_2 + \dots + a_nv_n = 0$ if and only if $a_1 = a_2 = \dots = a_n = 0$

A set of vectors is said to be **linearly dependent** if it is not linearly independent.

Intuitively, a set of vectors is linearly independent if each vector adds a new "direction", that cannot be accessed by any linear combination of the other vectors alone. In case of linearly dependent set, a vector in the set can be represented as a linear combination of the other vectors.

Basis of a vector space V is defined as a set of vectors that is linearly independent and spans

V .

Basis of a vector space is not unique. However, the number of vectors in the basis of a vector space remains the same for any given vector space. **Dimension** of a vector space is defined as the number of vectors in the basis of that vector space.

It can be proved that every vector in V can be written uniquely as a linear combination of the basis vectors.

A vector space is said to be **finite dimensional** if it has a finite dimension, i.e., a finite set of vectors is a basis for the space.

A vector space is said to be **infinite dimensional** if no finite set of vectors exists which is a basis for the space.

1.2 Linear Operators

In quantum mechanics, observables are represented by operators.

An **Operator** T , in general, is a function that takes a vector v from vector space V , to another vector $T(v)$ in another vector space W . If both the input and output spaces are the same, V , then T is said to be an operator on V .

A **Linear Operator** T on V , is a function that takes V to V , and has the following properties,

- $T(u + v) = T(u) + T(v)$, for all $u, v \in V$.
- $T(au) = aT(u)$, for all $a \in \mathbb{F}$, and $u \in V$.

The set of all linear operators on V is denoted by $\mathcal{L}(V)$.
Note that we define $Tu = T(u)$.

Natural definitions for addition and scalar multiplication of operators follow,

- $(T + S)v = Tv + Sv$ for all $T, S \in \mathcal{L}(V)$ and $v \in V$.
- $(aT)v = a(Tv)$ for all $a \in \mathbb{F}$, $T \in \mathcal{L}(V)$, $v \in V$.

Operators also have the operation of **multiplication**.

- $(ST)v = S(Tv)$, hence the product of two linear operators S and T is another linear operator ST .
- Operator multiplication is associative, i.e., $S(TU) = (ST)U$.
- Operator multiplication is, in general, **non-commutative**.

The **Commutator** of two operators is defined as $[T, S] = TS - ST$.

The **Anti-commutator** of two operators is defined as $\{T, S\} = TS + ST$.

The **Null Space** of an operator T is defined as the set of all vectors v that are mapped to the vector 0 by the operator.

$$\text{null}(T) = \{v : Tv = 0, v \in V\} \quad (1)$$

The **Range** of an operator T is defined as the image of V under T .

$$\text{range}(T) = \{Tv : v \in V\} \quad (2)$$

Note that both the null space and range of a linear operator are subspaces of the V .

An operator is said to be **injective** if $Tu = Tv$ if and only if $u = v$. It can be shown that T is injective if and only if $\text{null}(T) = 0$.

An operator is said to be **surjective** if $\text{range}(T) = V$.

Dimension theorem : The following is an important result from linear algebra,

$$\boxed{\dim V = \dim(\text{null}T) + \dim(\text{range}T)} \quad (3)$$

From this equation, we get that for finite-dimensional V , T is surjective if and only if T is injective.

Proof: Operator T is injective $\Rightarrow \dim(\text{null } T) = 0 \Rightarrow \dim(\text{range } T) = \dim V \Rightarrow$ since $\text{range } T \subseteq V$, we have $\text{range } T = V \Rightarrow T$ is surjective.

The inverse statement can be proved similarly.

Note - this does not hold true for infinite-dimensional V .

The **inverse** of an operator T is defined as another operator S such that $ST = TS = I$ (the identity operator).

An operator is said to be **invertible** if an inverse of that operator exists.

It can be proved that the inverse of an operator, if it exists, is unique.

Condition for invertibility -

An operator is invertible if and only if it is both injective and surjective.

From the previous result, we know that for finite-dimensional case, either one of injectivity or surjectivity is sufficient condition for invertibility, since any one implies the other.

However, for the infinite-dimensional case, both conditions must be satisfied.

Matrix Representation of a Linear Operator -

Till now, our discussion didn't rely on the choice of a specific basis, i.e., it was **basis-independent**.

However, for a finite-dimensional vector space V , if we choose a specific basis to work with, every abstract vector in the vector space can be represented as a column vector, and every linear operator T can be represented by a matrix.

Let the chosen basis be $e = (e_1, e_2, \dots, e_n)$. Note that e is not a vector, it a *set* of vectors.

We know that any vector v in the V can be represented uniquely as a linear combination of the basis vectors, $v = v_1e_1 + v_2e_2 + \dots + v_ne_n$. Hence, in this basis, the vector v has column vector

$$\text{representation } v \longleftrightarrow \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$$

Now, we also know that the action of a linear operator T on any arbitrary vector $v \in V$ is fixed once $Te_1, Te_2, \dots, Te_n$ are fixed, because then by linearity,

$$Tv = v_1Te_1 + v_2Te_2 + \dots + v_nTe_n \quad (4)$$

Now, consider a matrix $[T_{ij}]_{n \times n}$ whose columns are the column vector representations of $Te_1, Te_2, \dots, Te_n$.

$$Te_j \longleftrightarrow \begin{pmatrix} T_{1j} \\ T_{2j} \\ \vdots \\ T_{nj} \end{pmatrix} \quad (5)$$

which means that

$$Te_j = T_{1j}e_1 + T_{2j}e_2 + \dots + T_{nj}e_n \quad (6)$$

that is,

$$Te_j = \sum_{i=1}^n T_{ij}e_i \quad (7)$$

This matrix is defined as the matrix associated with the linear operator T .

Notational convention - In quantum formalism, in case subscript indices appear more than once in a product (like i in the equation (7)), often the sigma symbol is neglected and it is implicitly understood that the repeated index is summed over all its possible values. In this case, we may simply write equation (7) as -

$$Te_j = T_{ij}e_i \quad (8)$$

I will adopt this convention from now on for all our discussion. Unless explicitly stated otherwise, repeated indexes will be assumed to be summed over.

Now, in the given basis, Tv can be computed for any arbitrary vector v using usual matrix multiplication in the following way -

$$Tv \longleftrightarrow \begin{pmatrix} T_{11} & T_{12} & \dots & T_{1n} \\ T_{21} & T_{22} & \dots & T_{2n} \\ \vdots & \ddots & & \vdots \\ T_{n1} & T_{n2} & \dots & T_{nn} \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} \quad (9)$$

It is important to note that the column vector representation of abstract vectors, and matrix representation of linear operators depend on the particular choice of basis, i.e., they are **basis-dependent**. Hence, in some cases, it is useful to write the matrix representing T in basis $e = (e_1, e_2, \dots, e_n)$, explicitly as $T(\{e\})$. In some other basis f , the matrix representing T would be $T(\{f\})$, which would be different from $T(\{e\})$.

It can be shown that the matrix representation of the product of two operators ST is the matrix product of the matrices representing S and T , in that order.

The **trace** and **determinant** of operators are **basis-independent** quantities.

1.3 Eigenvalues and Eigenvectors

Eigenvalues and Eigenvectors are key concepts in quantum mechanics, because the outcome of any measurement is an eigenvalue of the observable's operator.

Consider the equation,

$$\boxed{Tv = \lambda v} \quad (10)$$

where $\lambda \in \mathbb{F}$, $v \in V$ and v is **nonzero**.

Here, λ is called an **eigenvalue** of the linear operator T and v is called an **eigenvector** of T corresponding to the eigenvalue λ .

Intuitively, eigenvectors of a linear operator are vectors in the vector space that are simply *scaled* in the same direction by a factor of λ , and are not rotated.

The set of eigenvectors of T is called the **spectrum** of T .

Equation (10) is equivalent to

$$(t - \lambda I)v = 0 \quad (11)$$

for some nonzero v .

From this, we get that

$$\text{Set of eigenvectors of } T \text{ corresponding to } \lambda = \text{null}(T - \lambda I) \quad (12)$$

Since v is a nonzero vector that belongs to $\text{null}(T - \lambda I)$, it implies that $(T - \lambda I)$ is not injective, which further implies that it is not invertible.

We can choose any basis and get the matrix representation of $(T - \lambda I)$. This matrix must be non-invertible (in any basis), which implies that

$$\boxed{\det(T - \lambda I) = 0} \quad (13)$$

This is called the **characteristic equation** of T , and $\det(T - \lambda I)$ is called the **characteristic polynomial** of T . The solutions of the characteristic equation give us all the eigenvalues of T .

From the Fundamental theorem of algebra, we can say that for a **complex** vector space of dimension n , any linear operator on it will have exactly n eigenvalues, allowing repeated roots. The case is different however, for a **real** vector space, where operators may have less than n eigenvalues, or may not have eigenvalues at all.

If all eigenvalues of T are distinct, the spectrum of T is said to be **non-degenerate**. If an eigenvalue appears k times, it is said to be a **degenerate** eigenvalue with multiplicity k . For a **complex** vector space, even in most degenerate case, there is at least one eigenvalue.

The following is an important theorem concerning eigenvectors -

If $\lambda_1, \lambda_2, \dots, \lambda_n$ are distinct eigenvalues of T and $v_1, v_2, \dots, v_n$ are the corresponding eigenvectors, then the set $(v_1, v_2, \dots, v_n)$ is linearly independent.

1.4 Inner Products

An **inner product** on a vector V over $\mathbb{F}$ is a map from an ordered pair (u, v) of vectors in V to a number $\langle u, v \rangle$ in $\mathbb{F}$, that satisfies the following properties -

- $\langle v, v \rangle \geq 0$, for all vectors $v \in V$.
- $\langle v, v \rangle = 0$ if and only if $v = 0$.
- $\langle u, v_1 + v_2 \rangle = \langle u, v_1 \rangle + \langle u, v_2 \rangle$ (Additivity in second entry).
- $\langle u, av \rangle = a\langle u, v \rangle$, for $a \in \mathbb{F}$ (Homogeneity in second entry).
- $\langle u, v \rangle = \langle v, u \rangle^*$ (Conjugate exchange symmetry).

A vector space with a definition for an inner product is called an **inner product space**.

Some additional properties of the inner product that may be derived using the above listed ones are -

- Combining additivity and homogeneity in second variable, we get that there is **linearity in second variable**, $\langle u, av_1 + bv_2 \rangle = a\langle u, v_1 \rangle + b\langle u, v_2 \rangle$, for $a, b \in \mathbb{F}$.

- Using conjugate exchange symmetry, we can show that there is **conjugate linearity in first variable**, $\langle au_1 + bu_2, v \rangle = a^* \langle u_1, v \rangle + b^* \langle u_2, v \rangle$

The inner product allows us to define the norm of a vector.

The **norm** of a vector $v \in V$ is defined as $|v| = \langle v, v \rangle$. The norm of any vector is a zero or a positive real quantity.

Vectors u and v are said to be **orthogonal** if and only if $\langle u, v \rangle = 0$.

The only vector that is orthogonal to all vectors in the space is the zero vector, i.e., if $\langle x, v \rangle = 0$ for all $v \in V$, then $x = 0$. We can see that by letting $v = 0$ and using second property of the inner product. This property is called the **non-degeneracy** of the inner product.

A set of vectors $(v_1, v_2, \dots, v_n)$ is called orthogonal if all the vectors are pairwise orthogonal.

A set of vectors $(v_1, v_2, \dots, v_n)$ is called **orthonormal** if it is orthogonal and all the vectors have norm one.

For an orthonormal set of vectors, we may concisely write

$$\boxed{\langle v_i, v_j \rangle = \delta_{ij}} \quad (14)$$

Pythagorean Theorem -

For any two orthogonal vectors $u, v \in V$,

$$|u + v|^2 = |u|^2 + |v|^2 \quad (15)$$

Schwarz inequality -

For any two vectors $u, v \in V$

$$|\langle u, v \rangle| \leq |u||v| \quad (16)$$

Triangle inequality -

For any two vectors $u, v \in V$

$$|u + v| \leq |u| + |v| \quad (17)$$

1.5 Orthonormal basis and Orthogonal projectors

It can be shown that the vectors in an orthonormal list are always linearly independent.

An **orthonormal basis** of V is an orthonormal set of vectors $(e_1, e_2, \dots, e_n)$ which is also a basis for V .

Then any vector $v \in V$ can be written uniquely as $v = v_1 e_1 + v_2 e_2 + \dots + v_n e_n$.

The constants v_i can be calculated as follows

$$\langle e_i, v \rangle = v_1 \langle e_i, e_1 \rangle + v_2 \langle e_i, e_2 \rangle + \dots + v_n \langle e_i, e_n \rangle = v_i \quad (18)$$

Therefore any vector v can be written as

$$v = \langle e_i, v \rangle e_i = \langle e_1, v \rangle e_1 + \langle e_2, v \rangle e_2 + \dots + \langle e_n, v \rangle e_n \quad (19)$$

Orthonormal basis are always preferred because working with them is easier and more convenient.

Any basis for a vector space V can be turned into an orthonormal basis using the **Gram-Schmidt Procedure** -

Given a list of vectors $(v_1, v_2, \dots, v_n)$ that is a non-orthonormal basis for V . We wish to generate an orthonormal basis $(e_1, e_2, \dots, e_n)$ for V . We can generate e_i recursively as follows -

If we know $e_1, e_2, \dots, e_{i-1}$, then we can write e_i as -

$$e_i = \frac{v_i - \langle e_1, v_i \rangle e_1 - \langle e_2, v_i \rangle e_2 - \dots - \langle e_{i-1}, v_i \rangle e_{i-1}}{|v_i - \langle e_1, v_i \rangle e_1 - \langle e_2, v_i \rangle e_2 - \dots - \langle e_{i-1}, v_i \rangle e_{i-1}|} \quad (20)$$

Intuitively, in this procedure, given vector v_i , we are deleting the components of v_i along $v_1, v_2, \dots, v_{i-1}$ and only retaining the orthogonal component. Then we divide by its norm in order to normalize it.

Let U be a subspace of V . Let $U^\perp$ be the **orthogonal complement subspace** of U , i.e., $U^\perp$ is the set of all vectors that are orthogonal to all the vectors in U .

For any vector $v \in V$, there is unique way to write $v = u + w$, where $u \in U$, and $w \in U^\perp$. In such case, a linear operator P_U called the **orthogonal projection** of V onto U , is defined, which maps v to u for every $v \in V$, and corresponding $u \in U$.

Properties of orthogonal projectors -

- $\text{range}(P_U) = U$
- $\text{null}(P_U) = U^\perp$
- If $(e_1, e_2, \dots, e_n)$ is an orthonormal basis for U , then for any vector $v \in V$, we have

$$P_U v = \langle e_1, v \rangle e_1 + \langle e_2, v \rangle e_2 + \dots + \langle e_n, v \rangle e_n \quad (21)$$

Intuitively, this means that we only add up the components of v that lie in the subspace U , and neglect other components.

- If we act on a vector v with P_U twice in succession, the second action has no effect as it acting on a vector already in U . This means

$$P_U^2 = P_U \quad (22)$$

- **Eigenvalues of projectors** - Equation (22) implies that if λ is an eigenvalue of P_U , then $\lambda^2 = \lambda$. This means that orthogonal projectors only have 0 and 1 as eigenvalues. The eigenvectors corresponding to 1 lie in U and the eigenvectors corresponding to 0 lie in $U^\perp$.

- **Determinant of projectors** - Clearly P_U is non-invertible which implies that

$$\det(P_U) = 0 \quad (23)$$

- **Trace of projectors** -

$$\text{tr } P_U = \dim U \quad (24)$$

Projectors are very useful in quantum mechanics. Measuring an observable on a physical state vector turns this original vector instantaneously into the orthogonal projection of the original vector down to some eigenspace of the operator associated with the observable.

1.6 Hermitian and Unitary Operators

Hermitian operators are fundamental to understanding quantum mechanics, because all observables are represented by Hermitian operators. Before discussing Hermitian operators, let us define the Hermitian conjugate of an operator.

The **adjoint** or **Hermitian conjugate** $T^\dagger$ of a linear operator T is defined as that operator which satisfies

$$\langle u, Tv \rangle = \langle T^\dagger u, v \rangle \quad (25)$$

for all vectors $u, v \in V$.

Properties of the Hermitian conjugate -

- It is **linear** operator.
- $(ST)^\dagger = T^\dagger S^\dagger$
- $(T^\dagger)^\dagger = T$
- The matrix representation of $T^\dagger$ is the **transpose of the complex conjugate** of the matrix representation of T , if and only if the chosen basis is **orthonormal**.

Proof:

Let us choose an orthonormal basis $(e_1, e_2, \dots, e_n)$. In equation (25), let us substitute u as e_i and v as e_j .

$$\langle e_i, Te_j \rangle = \langle e_i, T_{kj}e_k \rangle$$

From linearity of second variable and orthonormality condition,

$$\langle e_i, Te_j \rangle = T_{kj} \langle e_i, e_k \rangle = T_{kj} \delta_{ik} = T_{ij}$$

Also, from equation (25), and conjugate linearity of second variable,

$$\langle e_i, Te_j \rangle = \langle T^\dagger e_i, e_j \rangle = \langle T_{ki}^\dagger e_k, e_j \rangle = (T_{ki}^\dagger)^* \langle e_k, e_j \rangle = (T_{ki}^\dagger)^* \delta_{kj} = (T_{ji}^\dagger)^*$$

$$\text{Hence we get } T_{ij} = (T_{ji}^\dagger)^*$$

Hence proved.

Important Theorem -

Let T be a linear operator in **complex vector space** V , then

$$\text{if } \langle v, Tv \rangle = 0 \text{ for all } v \in V, \text{ then } T = 0 \quad (26)$$

An operator T is called **Hermitian** if $\boxed{T = T^\dagger}$.

Properties of Hermitian operators -

- The eigenvalues of Hermitian operators are always real.
- Different eigenvalues of a Hermitian operator correspond to orthogonal eigenvectors.
- T is Hermitian if and only if $\langle v, Tv \rangle \in \mathbb{R}$ for all v .

Now we discuss another extremely important class of operators - Unitary operators.

A linear operator U in a complex vector space V is called **Unitary** if it is surjective and does not change the norm of any vector that it acts on.

$$|Uv| = |v|, \text{ for all } v \in V \quad (27)$$

Properties of Unitary operators -

- null $U = \{0\}$, which implies that U is injective.
- Since U is both injective and surjective, it is invertible, even for infinite-dimensional vector spaces.
- The inverse of a unitary operator is its Hermitian conjugate.

$$\boxed{U^\dagger U = U U^\dagger = I} \quad (28)$$

- Unitary operators preserve inner products

$$\langle Uu, Uv \rangle = \langle u, v \rangle \quad (29)$$

1.7 Dirac's Bra-Ket Notation

Dirac introduced a new notational system for vectors, operators and inner products, to make working with them more convenient.

In this system, vector $v \in V$ is represented as $|v\rangle \in V$. This object is also known as a **ket**.

Sometimes, kets may be labelled by an index number. For example, the basis vectors $(e_1, e_2, \dots, e_n)$ may be written simply as $(|1\rangle, |2\rangle, \dots, |n\rangle)$.

The inner product $\langle u, v \rangle$ is written as $\langle u|v\rangle$. This notation is to be interpreted as $\langle u||v\rangle$, where the object $\langle u|$ is acting on the ket $|v\rangle \in V$. The object $\langle u|$ is called a **bra**, and is said to belong in V^* that is a dual space of V .

If we think of the column vector representation of ket $|u\rangle \in V$ as

$$|u\rangle \longleftrightarrow \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix} \quad (30)$$

then its dual element, bra $\langle u| \in V^*$ is the conjugate transpose of the column vector representation, i.e., it is a row vector of the form

$$\langle u| \longleftrightarrow (u_1^* \quad u_2^* \quad \dots \quad u_n^*) \quad (31)$$

and $\langle u|v\rangle$ is simply the matrix multiplication of $\langle u|$ and $|v\rangle$.

$$\langle u|v\rangle = u_1^* v_1 + u_2^* v_2 + \dots + u_n^* v_n \quad (32)$$

We can also define an **outer-product** using this notation.

$$|u\rangle\langle v| = \begin{pmatrix} u_1^* \\ u_2^* \\ \vdots \\ u_n^* \end{pmatrix} (v_1 \quad v_2 \quad \dots v_n) = \begin{pmatrix} u_1^* v_1 & u_1^* v_2 & \dots & u_1^* v_n \\ u_2^* v_1 & u_2^* v_2 & \dots & u_2^* v_n \\ \vdots & \ddots & & \vdots \\ u_n^* v_1 & u_n^* v_2 & \dots & u_n^* v_n \end{pmatrix} \quad (33)$$

Clearly, the outer-product is a matrix. Hence, it can be interpreted as an operator.

$$|a\rangle\langle b||v\rangle = |a\rangle\langle b|v\rangle \quad (34)$$

Here, $|a\rangle\langle b|$ is acting as an operator on vector $|v\rangle$ to return another vector $|a\rangle\langle b|v\rangle$, which is simply $|a\rangle$ (a vector) scaled by $\langle b|v\rangle$ (a number).

A familiar example of an outer product acting as an operator is this -

Consider any vector $|v\rangle \in V$, and let $(|1\rangle, |2\rangle, \dots, |n\rangle)$ be the basis vectors. Then writing $|v\rangle$ as a linear combination of the basis vectors, we get

$$|v\rangle = v_i |i\rangle \quad (35)$$

But we also know that

$$v_i = \langle i|v\rangle \quad (36)$$

From this, we get that

$$|v\rangle = |i\rangle\langle i|v\rangle \quad (37)$$

Hence

$$\boxed{|i\rangle\langle i| = I, \text{ the identity operator}} \quad (38)$$

This property is called the **resolution of the identity**. Keep in mind that the index i is being summed over here.

Individual terms, $|i\rangle\langle i|$ (i not summed) represent **projection operators** along the basis vector $|i\rangle$.

The action of a **linear operator** T on a ket $|v\rangle \in V$ is denoted as $T|v\rangle$. It is also sometimes denoted as $|Tv\rangle$.

We can derive an expression the matrix elements of T in some orthonormal basis $(|1\rangle, |2\rangle, \dots, |n\rangle)$. Assume $|b\rangle$ is obtained by the action of T on $|a\rangle$.

$$T|a\rangle = |b\rangle \quad (39)$$

$$T|n\rangle a_n = |n\rangle b_n \quad (40)$$

Operating by $\langle m|$ on both sides, and using orthonormality condition,

$$\langle m|T|n\rangle a_n = \langle m|n\rangle b_n = b_n \delta_{mn} = b_m \quad (41)$$

Comparing with the matrix multiplication rule, we get that

$$\boxed{T_{ij} = \langle i|T|j\rangle} \quad (42)$$

The following can be proved for operator T ,

$$\langle u|T^\dagger|v\rangle = \langle v|T|u\rangle^*, \text{ for all } u, v \quad (43)$$

1.8 Hilbert Spaces

A **Hilbert space** is an **inner product space** that is **complete**. **Completeness** means that all Cauchy sequences of vectors from the space, converge to a vector that is also in the space.

The Hilbert space $\mathcal{H}$ that is dealt with in quantum mechanics is the space of all **square-integrable** functions $\Psi(x, y, z, t)$.

We can define a basis for this space in the **position-basis** as

$$|x_0\rangle = \delta(x - x_0), \text{ for all } x_0 \in \mathbb{R} \quad (44)$$

This basis vector represents a particle perfectly localized at the position x_0 .

Note that here, x_0 is distinct from the vector $|x_0\rangle$. In fact, x_0 is only a label. Hence, $|ax_0\rangle \neq a|x_0\rangle$.

Since, it is an infinite-dimensional space and x_0 can take continuous values, it is called a **non-denumerable basis**.

We define the inner product of basis vectors as

$$\langle x|y\rangle = \delta(x - y) \quad (45)$$

From this definition, we satisfy a **completeness relation**, which is nothing but the resolution of the identity, analogous to equation (38)

$$\int |x\rangle\langle x|dx = 1, \text{ the identity operator} \quad (46)$$

If the state of the particle is $|\psi\rangle$, then we define $\psi(x)$ to be its projection along the basis vector $|x\rangle$. Hence, we have

$$\boxed{\psi(x) = \langle x|\psi\rangle \in \mathbb{C}} \quad (47)$$

From these definitions, we also get the expression for the inner product for two arbitrary state vectors $\phi(x)$ and $\psi(x)$ as -

$$\langle \phi|\psi\rangle = \langle \phi|1|\psi\rangle \quad (48)$$

Resolving the identity, we get

$$\langle \phi|\psi\rangle = \int \langle \phi|x\rangle\langle x|\psi\rangle dx = \int \langle x|\phi\rangle^* \langle x|\psi\rangle dx \quad (49)$$

Hence,

$$\boxed{\langle \phi|\psi\rangle = \int \phi^*(x)\psi(x)dx} \quad (50)$$

The **position operator** is defined by its action on position basis vectors.

$$\hat{x}|x\rangle = x|x\rangle \quad (51)$$

Completely similar to the position space, basis vectors in the momentum space can be defined as $|p\rangle$, which represents a completely delocalized plane wave with momentum p . All properties are similar as the position-space case.

$$\langle p'|p\rangle = \delta(p' - p) \quad (52)$$

$$\int |p'\rangle\langle p|dp = 1 \quad (53)$$

$$\hat{p}|p\rangle = p|p\rangle \quad (54)$$

We can also compute the inner product of a position basis state and a momentum basis state, from wave mechanics

$$\langle x|p\rangle = \frac{e^{ipx/\hbar}}{\sqrt{2\pi\hbar}} \quad (55)$$

We can also compute the component of a general state ψ along the position basis vector $|p\rangle$

$$\langle p|\psi\rangle = \int \langle p|x\rangle\langle x|\psi\rangle dx = \frac{1}{\sqrt{2\pi\hbar}} \int e^{-ipx/\hbar}\psi(x)dx = \tilde{\psi}(p) \quad (56)$$

This is nothing but the Fourier transform of $\psi(x)$, which we already know from wave mechanics, takes us from the position to the momentum basis.

2 The Uncertainty Principle

2.1 Expectation and Uncertainty of Operators

The **expectation value** of the operator Q on the state $|\Psi\rangle$ is given by

$$\boxed{\langle Q \rangle = \langle \Psi | Q | \Psi \rangle} \quad (57)$$

The **uncertainty** of a Hermitian operator Q in a state $|\Psi\rangle$ is defined as

$$\boxed{\Delta Q(\Psi) = |(Q - \langle Q \rangle I) \Psi|} \quad (58)$$

Some properties of the uncertainty so defined are -

- The uncertainty is a non-negative real number.
- The uncertainty in Q is zero if and only if $|\Psi\rangle$ is an eigenstate of Q .
- The following alternative expression of uncertainty can be derived

$$\boxed{(\Delta Q(\Psi))^2 = \langle Q^2 \rangle - \langle Q \rangle^2} \quad (59)$$

2.2 The Uncertainty Principle

Consider two Hermitian operators A and B on a quantum system in state $|\Psi\rangle$. Let ΔA and ΔB denote the uncertainties in these operators in this state. Then we have

$$\boxed{(\Delta A)^2(\Delta B)^2 \geq (\langle \Psi | \frac{1}{2i} [A, B] | \Psi \rangle)^2} \quad (60)$$

Proof:

Define the following two states:

$$|f\rangle = (A - \langle A \rangle I) |\Psi\rangle \quad (61)$$

$$|g\rangle = (B - \langle B \rangle I) |\Psi\rangle \quad (62)$$

Note that by definition of uncertainty, $\langle f | f \rangle = (\Delta A)^2$ and $\langle g | g \rangle = (\Delta B)^2$.

The Schwarz inequality gives us

$$\langle f | f \rangle \langle g | g \rangle \geq |\langle f | g \rangle|^2 \quad (63)$$

and therefore we have

$$(\Delta A)^2(\Delta B)^2 \geq |\langle f | g \rangle|^2 = (\text{Re}\langle f | g \rangle)^2 + (\text{Im}\langle f | g \rangle)^2 \quad (64)$$

Writing $\check{A} = (A - \langle A \rangle I)$ and $\check{B} = (B - \langle B \rangle I)$, we get

$$\langle f | g \rangle = \langle \Psi | \check{A} \check{B} | \Psi \rangle = \langle \Psi | (A - \langle A \rangle I)(B - \langle B \rangle I) | \Psi \rangle = \langle \Psi | AB | \Psi \rangle - \langle A \rangle \langle B \rangle \quad (65)$$

Similarly, we also get

$$\langle g | f \rangle = \langle \Psi | BA | \Psi \rangle - \langle B \rangle \langle A \rangle \quad (66)$$

From the above two equations, we get the imaginary part of $\langle f | g \rangle$ as

$$\text{Im}\langle f | g \rangle = \frac{1}{2i}(\langle f | g \rangle - \langle g | f \rangle) = \frac{1}{2i}\langle \Psi | [A, B] | \Psi \rangle \quad (67)$$

We can also compute the real part as

$$\text{Re}\langle f|g\rangle = \frac{1}{2}(\langle f|g\rangle + \langle g|f\rangle) = \frac{1}{2}\langle\Psi|\{\check{A}, \check{B}\}|\Psi\rangle \quad (68)$$

Putting these values back in equation (64), we get

$$(\Delta A)^2(\Delta B)^2 \geq \left(\frac{1}{2i}\langle\Psi|[A, B]|\Psi\rangle\right)^2 + \left(\frac{1}{2}\langle\Psi|\{\check{A}, \check{B}\}|\Psi\rangle\right)^2 \quad (69)$$

In this equation, the second term is either zero or a positive real, so it can be dropped while preserving the inequality. This gives us the famous uncertainty relation.

Hence proved.

Hence, the uncertainty principle tells us that we cannot measure simultaneously two observables whose operators don't commute, both with perfect accuracy. Such observables whose operators don't commute are called **incompatible observables**. A classic example of incompatible observables is position-momentum. The uncertainty principle for position and momentum comes to be of the form

$$\Delta x \Delta p \geq \frac{\hbar}{2} \quad (70)$$

2.3 Diagonalization and Normal Operators

We know that the matrix representation of an operator is basis-dependent.

If there exists a basis in which an operator T is represented by a diagonal matrix, then T is said to be **diagonalizable**.

It can be shown that in such cases, the basis vectors are eigenvectors of the operator T . Hence we conclude, that an operator is diagonalizable if and only if it possesses a set of eigenvectors that **span** the space.

Moreover, if there exists an **orthonormal** basis in which the matrix representation of T is diagonal, T is called **unitarily diagonalizable**.

Hence, an operator T is unitarily diagonalizable if and only if it has an orthonormal set of eigenvectors that span the space.

An operator is called **normal** if it commutes with its Hermitian conjugate, i.e.,

$$M \text{ is normal: } [M^\dagger, M] = 0 \quad (71)$$

Hermitian, anti-Hermitian and unitary operators are all examples of normal operators.

The following important theorem can be proved -

Spectral theorem -

An operator M in a complex vector space is unitarily diagonalizable if and only if M is normal.

Two operators S and T are called **simultaneously diagonalizable** if there exists a basis in which the matrix representation of both of these operators are diagonal.

The following two theorems can be proved -

Theorem 1: If S and T are simultaneously diagonalizable, then S and T commute.

Theorem 2: If S and T are both Hermitian operators that commute, then S and T are simultaneously diagonalizable.

This means that **incompatible observables** *cannot* be simultaneously diagonalized.

On the other hand, **compatible observables** *can* be simultaneously diagonalized.

3 The Postulates of Quantum Mechanics

Postulate 1: The State of a System

The state of a quantum system is completely specified, at each time t , by a state vector $|\Psi(t)\rangle$ which belongs in a Hilbert space.

Postulate 2: Observables and operators

To every physically measurable quantity Q , called an observable, corresponds a Hermitian operator $\hat{Q}$, whose eigenvectors form a complete basis. The expectation value of Q in the state $|\Psi\rangle$ is given by $\langle\Psi|\hat{Q}|\Psi\rangle$.

Postulate 3: Measurements and eigenvalues of operators

When a measurement of the observable Q is made on a quantum system in state $|\Psi\rangle$, the only possible outcomes of the measurement are the eigenvalues, Q_n , of the operator $\hat{Q}$.

$$\hat{Q}|\Psi_n\rangle = Q_n|\Psi_n\rangle \quad (72)$$

The effect of making the measurement is to *instantaneously* collapse the state of the system $|\Psi\rangle$ to the corresponding eigenstate of the system $|\Psi_n\rangle$ after the measurement.

Postulate 4: The Generalized Statistical Interpretation

Upon making a measurement of the observable Q on a quantum system in state $|\Psi\rangle$, the probability of getting a particular eigenvalue Q_n as the outcome is equal to the square of the component of $|\Psi\rangle$ along the eigenstate $|\Psi_n\rangle$.

For **discrete spectra**, probability of obtaining a non-degenerate eigenvalue Q_n is given by

$$P(Q_n) = \frac{|\langle\Psi_n|\Psi\rangle|^2}{\langle\Psi|\Psi\rangle} \quad (73)$$

For an eigenvalue with degeneracy m , the probability of obtaining it is given by

$$P(Q_n) = \frac{\sum_{j=1}^m |\langle\Psi_n^j|\Psi\rangle|^2}{\langle\Psi|\Psi\rangle} \quad (74)$$

For **continuous spectra**, if q is an eigenvalue of the observable, then the probability of measuring an outcome between q and $q+dq$ is given by

$$\frac{dP(q)}{dq} = \frac{|\Psi(q)|^2}{\langle\Psi|\Psi\rangle} \quad (75)$$

Postulate 5: Time Evolution of a Quantum System

The time evolution of the state $|\Psi(t)\rangle$ of a quantum system is governed by the **Schrodinger equation**,

$$\boxed{i\hbar \frac{\partial |\Psi(t)\rangle}{\partial t} = \hat{H}|\Psi(t)\rangle} \quad (76)$$

where $\hat{H}$ is the **Hamiltonian** operator corresponding to the total energy of the system.

We shall consider the time evolution of a quantum system in greater detail in the following section.

4 Quantum Dynamics

The state space of quantum mechanics, which is a Hilbert space $\mathcal{H}$, is a space of **time-independent** basis vectors. Time is a separate parameter; at different times, the state of the quantum system is described by different state vectors in $\mathcal{H}$. Thus, we write the state vector at time t as $|\Psi, t\rangle$.

Since the state must be normalized, we can imagine the state $|\Psi, t\rangle$ as a unit vector whose tip sweeps a trajectory in $\mathcal{H}$ as a function of time.

4.1 Unitary Time Evolution

We know that unitary operators are linear operators that preserve the norm of any vector that they act upon. Since the state vector must remain normalized as it evolves through time, we postulate the following -

We declare that for a given quantum system, there exists a unique **unitary** operator $\mathcal{U}(t, t_0)$ such that for *any* initial state $|\Psi, t_0\rangle$, the state at time t is given by -

$$\boxed{|\Psi, t\rangle = \mathcal{U}(t, t_0)|\Psi, t_0\rangle, \forall t, t_0} \quad (77)$$

Note that the *same* operator $\mathcal{U}(t, t_0)$ takes *any* initial state $|\Psi, t_0\rangle$ to its corresponding state $|\Psi, t\rangle$ at time t . A physical system has a single operator $\mathcal{U}$ that generates the time evolution of all possible initial states.

Properties of $\mathcal{U}(t, t_0)$ -

- If we set $t = t_0$, we must get the identity operator, i.e.,

$$\mathcal{U}(t_0, t_0) = 1, \forall t \quad (78)$$

- If we consider time evolution from t_0 to t_1 , and then from t_1 to t_2 , we get that

$$\mathcal{U}(t_2, t_0) = \mathcal{U}(t_2, t_1)\mathcal{U}(t_1, t_0) \quad (79)$$

- If we set $t_2 = t_0$ in the previous result, we get that

$$\mathcal{U}(t_0, t) = \mathcal{U}^{-1}(t, t_0) = \mathcal{U}^\dagger(t, t_0) \quad (80)$$

4.2 Schrodinger Dynamics

The Schrodinger equation can be derived from the unitary time evolution postulate. In that sense, it is only an alternative way to say the statement of unitary time evolution.

Take the time derivative of the unitary time evolution equation (77)

$$\frac{\partial}{\partial t}|\Psi, t\rangle = \frac{\partial \mathcal{U}(t, t_0)}{\partial t}|\Psi, t_0\rangle \quad (81)$$

Substitute $|\Psi, t_0\rangle$ in the right hand side,

$$\frac{\partial}{\partial t}|\Psi, t\rangle = \frac{\partial \mathcal{U}(t, t_0)}{\partial t}\mathcal{U}(t_0, t)|\Psi, t\rangle = \frac{\partial \mathcal{U}(t, t_0)}{\partial t}\mathcal{U}^\dagger(t, t_0)|\Psi, t\rangle \quad (82)$$

Let us define

$$\Lambda(t, t_0) = \frac{\partial \mathcal{U}(t, t_0)}{\partial t}\mathcal{U}^\dagger(t, t_0) \quad (83)$$

so that

$$\frac{\partial}{\partial t}|\Psi, t\rangle = \Lambda(t, t_0)|\Psi, t\rangle \quad (84)$$

Note that

$$\Lambda^\dagger(t, t_0) = \mathcal{U}(t, t_0) \frac{\partial \mathcal{U}^\dagger(t, t_0)}{\partial t} \quad (85)$$

We will prove two important properties of $\Lambda(t, t_0)$ -

- $\Lambda(t, t_0)$ is anti-Hermitian.

Proof:

Begin with the equation

$$\mathcal{U}(t, t_0)\mathcal{U}^\dagger(t, t_0) = 1 \quad (86)$$

Take a time-derivative to get

$$\frac{\partial \mathcal{U}(t, t_0)}{\partial t} \mathcal{U}^\dagger(t, t_0) + \mathcal{U}(t, t_0) \frac{\partial \mathcal{U}^\dagger(t, t_0)}{\partial t} = 0 \quad (87)$$

which, from equations (83) and (85), implies

$$\Lambda(t, t_0) + \Lambda^\dagger(t, t_0) = 0 \quad (88)$$

which proves that $\Lambda(t, t_0)$ is anti-Hermitian.

- $\Lambda(t, t_0)$ is actually independent of t_0 .

Proof:

To prove this, we will show that $\Lambda(t, t_0) = \Lambda(t, t_1)$ for any t_1 other than t_0 .

Let us start by inserting the identity operator in the expression for $\Lambda(t, t_0)$

$$\begin{aligned} \Lambda(t, t_0) &= \frac{\partial \mathcal{U}(t, t_0)}{\partial t} \mathcal{U}^\dagger(t, t_0) \\ &= \frac{\partial \mathcal{U}(t, t_0)}{\partial t} (\mathcal{U}(t_0, t_1) \mathcal{U}^\dagger(t_0, t_1)) \mathcal{U}^\dagger(t, t_0) \end{aligned} \quad (89)$$

Since $\mathcal{U}(t_0, t_1)$ is independent of t ,

$$\begin{aligned} \Lambda(t, t_0) &= \frac{\partial}{\partial t} (\mathcal{U}(t, t_0) \mathcal{U}(t_0, t_1)) \mathcal{U}^\dagger(t_0, t_1) \mathcal{U}^\dagger(t, t_0) \\ &= \frac{\partial \mathcal{U}(t, t_1)}{\partial t} \mathcal{U}(t_1, t_0) \mathcal{U}(t_0, t) = \frac{\partial \mathcal{U}(t, t_1)}{\partial t} \mathcal{U}(t_1, t) \\ &= \frac{\partial \mathcal{U}(t, t_1)}{\partial t} \mathcal{U}^\dagger(t, t_1) = \Lambda(t, t_1) \end{aligned} \quad (90)$$

Hence proved.

Hence we simply write $\Lambda(t) = \Lambda(t, t_0)$.

From this, equation (84) becomes

$$\frac{\partial}{\partial t}|\Psi, t\rangle = \Lambda(t)|\Psi, t\rangle \quad (91)$$

Define an operator $H(t)$ as

$$H(t) = i\hbar\Lambda(t) = i\hbar \frac{\partial \mathcal{U}(t, t_0)}{\partial t} \mathcal{U}^\dagger(t, t_0) \quad (92)$$

$H(t)$ has units of energy. Moreover, since $\Lambda(t)$ is anti-Hermitian, we get that $H(t)$ is Hermitian. We call it the **Hamiltonian operator** and it corresponds to the total energy of the system.

Multiplying equation(91) by $i\hbar$, we get the Schrodinger equation!

$$\boxed{\text{Schrodinger equation: } i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle} \quad (93)$$

4.3 Heisenberg Dynamics

The operators we had been discussing so far (so called 'Schrodinger operators') may or may not have time-dependence. The idea of introducing 'Heisenberg operators' is to include the dynamical evolution into the operators themselves, rather than the state.

For every Schrodinger operator, we can define a corresponding Heisenberg operator. Let us consider a Schrodinger operator $\hat{A}_S$. Then its Heisenberg operator is defined as -

$$\hat{A}_H(t) = \mathcal{U}^\dagger(t, 0) \hat{A}_S \mathcal{U}(t, 0) \quad (94)$$

where $\mathcal{U}(t, 0)$ is the unitary time evolution operator as discussed in the previous sections.

It can be shown that the Heisenberg operator associated with the product of two Schrodinger operators is equal to the product of the corresponding Heisenberg operators, i.e.,

$$\hat{C}_S = \hat{A}_S \hat{B}_S \implies \hat{C}_H = \hat{A}_H \hat{B}_H$$

Expectation Values -

The expectation value of the Schrodinger operator $\hat{A}_S(t)$ is given by -

$$\begin{aligned} \langle \Psi, t | \hat{A}_S(t) | \Psi, t \rangle &= \langle \Psi, t | \mathcal{U}(t, 0) \hat{A}_H(t) \mathcal{U}^\dagger(t, 0) | \Psi, t \rangle \\ &= \langle \Psi, t | \mathcal{U}^\dagger(0, t) \hat{A}_H(t) \mathcal{U}(0, t) | \Psi, t \rangle \\ &= \langle \Psi, 0 | \hat{A}_H(t) | \Psi, 0 \rangle \end{aligned}$$

Hence we have -

$$\langle \Psi, t | \hat{A}_S(t) | \Psi, t \rangle = \langle \Psi, 0 | \hat{A}_H(t) | \Psi, 0 \rangle \quad (95)$$

We also write this as -

$$\langle \hat{A}_S \rangle = \langle \hat{A}_H(t) \rangle \quad (96)$$

Heisenberg Equation of Motion - The Heisenberg operators themselves satisfy a differential equation known as the Heisenberg equation of motion. Let us derive it.

From the definition of the Hamiltonian (equation 92), we have -

$$i\hbar \frac{\partial \mathcal{U}(t, t_0)}{\partial t} = H_S(t) \mathcal{U}(t, t_0) \quad (97)$$

Taking its adjoint, we have -

$$i\hbar \frac{\partial \mathcal{U}^\dagger(t, t_0)}{\partial t} = -\mathcal{U}^\dagger(t, t_0) H_S(t) \quad (98)$$

Now, we take the time derivative of equation (94), to get -

$$i\hbar \frac{d}{dt} \hat{A}_H(t) = \left(i\hbar \frac{\partial \mathcal{U}^\dagger}{\partial t}(t, 0) \right) \hat{A}_S(t) \mathcal{U}(t, 0) + \mathcal{U}^\dagger(t, 0) \hat{A}_S(t) \left(i\hbar \frac{\partial \mathcal{U}}{\partial t}(t, 0) \right) + \mathcal{U}^\dagger(t, 0) i\hbar \frac{\partial \hat{A}_S(t)}{\partial t} \mathcal{U}(t, 0)$$

Substituting from equations (97) and (98), we get -

$$i\hbar \frac{d}{dt} \hat{A}_H(t) = -\mathcal{U}^\dagger(t, 0) H_S(t) \hat{A}_S(t) \mathcal{U}(t, 0) + \mathcal{U}^\dagger(t, 0) \hat{A}_S(t) H_S(t) \mathcal{U}(t, 0) + \mathcal{U}^\dagger(t, 0) i\hbar \frac{\partial \hat{A}_S(t)}{\partial t} \mathcal{U}(t, 0)$$

Note that the last term is simply the Heisenberg operator associated with $i\hbar \frac{\partial \hat{A}_S(t)}{\partial t}$. Hence, we have -

$$i\hbar \frac{d}{dt} \hat{A}_H(t) = -H_H(t) \hat{A}_H(t) + \hat{A}_H(t) H_H(t) + i\hbar \left(\frac{\partial \hat{A}_S(t)}{\partial t} \right)_H$$

Note that the first two terms can be written as a commutator. Hence we get -

$$\boxed{i\hbar \frac{d}{dt} \hat{A}_H(t) = [\hat{A}_H(t), H_H(t)] + i\hbar \left(\frac{\partial \hat{A}_S(t)}{\partial t} \right)_H} \quad (99)$$

This is the **Heisenberg equation of motion**.

5 Quantum Mechanics in 3-Dimensions

5.1 Schrodinger's Equation in 3-Dimensions

We know that the Schrodinger's equation is -

$$i\hbar \frac{\partial \Psi}{\partial t} = H \Psi$$

In three dimensions, the Hamiltonian H is given as -

$$H = \frac{p^2}{2m} + V = \frac{1}{2m} (p_x^2 + p_y^2 + p_z^2) + V \quad (100)$$

where the momentum operators are defined as -

$$p_x = \frac{\hbar}{i} \frac{\partial}{\partial x}, \quad p_y = \frac{\hbar}{i} \frac{\partial}{\partial y}, \quad p_z = \frac{\hbar}{i} \frac{\partial}{\partial z}$$

So, we have -

$$\mathbf{p} = \frac{\hbar}{i} \nabla \quad (101)$$

Hence, we get the Schrodinger equation in three dimensions as -

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \Psi + V \Psi \quad (102)$$

where, ∇^2 is the Laplacian operator.

The probability of finding the particle in the infinitesimal volume element $d^3\mathbf{r} = dxdydz$ is

$$|\Psi(\mathbf{r}, t)|^2 d^3\mathbf{r}.$$

The normalization condition can be stated as -

$$\int |\Psi(\mathbf{r}, t)|^2 d^3\mathbf{r} = 1 \quad (103)$$

where the integral is taken over all space.

For time-independent potentials, we can separate the time and space variables in Ψ , exactly similar to the 1-dimensional case.

$$\Psi_n(\mathbf{r}, t) = \psi_n(\mathbf{r}) e^{-iE_n t/\hbar} \quad (104)$$

The spatial part satisfies the **time-independent Schrodinger equation** -

$$-\frac{\hbar^2}{2m} \nabla^2 \psi_n + V \psi_n = E_n \psi_n \quad (105)$$

The complete solution is the superposition of the eigensolutions as -

$$\Psi(\mathbf{r}, t) = \sum c_n \psi_n(\mathbf{r}) e^{-iE_n t/\hbar}$$

5.2 Schrodinger's Equation in Spherical Coordinates

For spherically-symmetric potentials, it becomes very convenient to work in spherical coordinates.

In spherical coordinates, the Laplacian takes the form -

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \left(\frac{\partial^2}{\partial \phi^2} \right)$$

Putting this in equation (102), we get -

$$-\frac{\hbar^2}{2m} \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \left(\frac{\partial^2 \psi}{\partial \phi^2} \right) \right] + V \psi = E \psi \quad (106)$$

We try separation of variables,

$$\psi(r, \theta, \phi) = R(r) Y(\theta, \phi) \quad (107)$$

Putting this into equation (106) and simplifying, we get,

$$\left\{ \frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) - \frac{2mr^2}{\hbar^2} [V(r) - E] \right\} + \frac{1}{Y} \left\{ \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial Y}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2 Y}{\partial \phi^2} \right\} = 0$$

Note that the first term is only dependent on r , whereas the second is only dependent on θ and ϕ . Hence both terms must be constant. We let -

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) - \frac{2mr^2}{\hbar^2} [V(r) - E] = l(l+1) \quad (108)$$

$$\frac{1}{Y} \left\{ \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial Y}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2 Y}{\partial \phi^2} \right\} = -l(l+1) \quad (109)$$

Here, l is called the **Azimuthal quantum number**.

Solving for the Angular part -

Let us try to find the angular part of the wave function by solving equation (109). We again employ separation of variables,

$$Y(\theta, \phi) = \Theta(\theta)\Phi(\phi)$$

Putting this into equation (109) and simplifying, we get -

$$\left\{ \frac{1}{\Theta} \left[\sin\theta \frac{d}{d\theta} \left(\sin\theta \frac{d\Theta}{d\theta} \right) \right] + l(l+1)\sin^2\theta \right\} + \frac{1}{\Phi} \frac{d^2\Phi}{d\phi^2} = 0$$

Note again that the first term is dependent only on θ , while the second term is dependent only on ϕ . Hence they are constants.

$$\frac{1}{\Theta} \left[\sin\theta \frac{d}{d\theta} \left(\sin\theta \frac{d\Theta}{d\theta} \right) \right] + l(l+1)\sin^2\theta = m^2 \quad (110)$$

$$\frac{1}{\Phi} \frac{d^2\Phi}{d\phi^2} = -m^2 \quad (111)$$

Here, m is called the **magnetic quantum number**.

The solution of equation (110) is given by -

$$\Theta(\theta) = AP_{l,m}(\cos\theta) \quad (112)$$

where $P_{l,m}$ is the **associated Legendre function**, defined by -

$$P_{l,m}(x) = (1-x^2)^{|m|/2} \left(\frac{d}{dx} \right)^{|m|} P_l(x)$$

where $P_l(x)$ is the l th **Legendre polynomial**.

For this to make sense, l must be a non-negative integer. Moreover, for any given value of l , m can take $(2l+1)$ different values, which are $m = -l, -l+1, \dots, -1, 0, 1, \dots, l-1, l$. We will see a proof of this statement in section (6.2).

The solution of equation (111) is -

$$\Phi(\phi) = e^{im\phi} \quad (113)$$

Equations (112) and (113) give us the angular part $Y_{l,m}(\theta, \phi) = \Theta(\theta)\Phi(\phi)$ of the wave functions. The normalized angular wave functions are called **spherical harmonics**.

It can be shown that the spherical harmonics constitute an **orthonormal set**.

Solving for the Radial part -

Define -

$$u(r) = rR(r) \quad (114)$$

Putting this into equation (108) and simplifying, we get -

$$\boxed{-\frac{\hbar^2}{2m} \frac{d^2u}{dr^2} + \left[V + \frac{\hbar^2}{2m} \frac{l(l+1)}{r^2} \right] u = Eu} \quad (115)$$

This is called the **radial equation**. It is identical to the one-dimensional Schrodinger equation, with an effective potential -

$$V_{\text{eff}} = V + \frac{\hbar^2}{2m} \frac{l(l+1)}{r^2} \quad (116)$$

The additional term $(\hbar^2/2m)[l(l+1)/r^2]$ is called the **Centrifugal** term.

Now, we apply the above developed theory to solve some simple 3-D systems with spherically symmetric potentials.

5.3 The Free Particle

In case of the free particle, the potential $V(\mathbf{r})$ is zero everywhere in space. Hence, the radial equation takes the form -

$$\frac{d^2u}{dr^2} = \left[\frac{l(l+1)}{r^2} - k^2 \right] u \quad (117)$$

where

$$k = \frac{\sqrt{2mE}}{\hbar}$$

The solution to this differential equation, that shows regular behaviour at the origin, is given by -

$$u(r) = Ar j_l(kr) \quad (118)$$

where $j_l(x)$ is the **spherical Bessel function** of order l .

5.4 The Infinite Spherical Well

The infinite spherical well is characterized by this potential -

$$V(r) = \begin{cases} 0, & \text{if } r < a \\ \infty, & \text{if } r > a \end{cases}$$

Outside the well, the wave function is zero.

Inside the well, it follows equation (117), subject to boundary condition $u(a) = 0$

In the case $l = 0$, we simply have -

$$\frac{d^2u}{dr^2} = -k^2u \implies u(r) = A\sin(kr) + B\cos(kr)$$

In order to avoid a δ function term in the effective potential, we set $B = 0$.

The boundary condition then requires that $\sin(ka) = 0$, and hence $ka = n\pi$, for some integer n . Hence, we get the energy eigenvalues -

$$E_{n,0} = \frac{n^2\pi^2\hbar^2}{2ma^2} \quad (119)$$

which is the same result as the 1-D infinite square well.

Normalization gives us $A = \sqrt{2/a}$.

Including the angular part $Y_{0,0}(\theta, \phi) = 1/\sqrt{4\pi}$ as well, we can write the full wave function for $l = 0$ as -

$$\Psi_{n,0,0} = \frac{1}{\sqrt{2\pi a}} \frac{\sin(n\pi r/a)}{r} \quad (120)$$

The stationary state $\Psi_{n,l,m}(r, \theta, \phi)$ is labelled by three quantum numbers n , l and m . Here n is called the **principal quantum number**. Note that the energy $E_{n,l}$ only depends on n and l and not on m .

For arbitrary integer l , we know from our discussion on the free particle that the wave function inside the sphere has the form $R(r) = A j_l(kr)$.

The boundary condition requires that $R(a) = 0$, which implies $j_l(ka) = 0$. Let $\beta_{n,l}$ denote the n th zero of the l th spherical Bessel function. These zeros can be calculated numerically.

Then, we have,

$$k = \frac{\beta_{n,l}}{a} \quad (121)$$

Hence, the allowed energies are -

$$E_{n,l} = \frac{\hbar^2}{2ma^2} \beta_{n,l}^2 \quad (122)$$

The wave functions are -

$$\Psi_{n,l,m}(r, \theta, \phi) = A_{n,l} j_l(\beta_{n,l} r/a) Y_{l,m}(\theta, \phi) \quad (123)$$

where the constant $A_{n,l}$ is to be determined from normalization.

5.5 The Hydrogen Atom

The Hydrogen atom consists of a proton placed at the origin, and an electron experiencing an attractive Coulomb force due to the proton. The proton may be assumed to be motionless because of its high mass compared to the electron. The potential energy has the form -

$$V(r) = -\frac{e^2}{4\pi\epsilon_0} \frac{1}{r} \quad (124)$$

The radial equation takes the form -

$$-\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} + \left[-\frac{e^2}{4\pi\epsilon_0} \frac{1}{r} + \frac{\hbar^2}{2m} \frac{l(l+1)}{r^2} \right] u = Eu \quad (125)$$

with $E < 0$, which represents a bound state.

Define -

$$\kappa = \frac{\sqrt{-2mE}}{\hbar} \quad (126)$$

$$\rho = \kappa r \quad (127)$$

$$\rho_0 = \frac{me^2}{2\pi\epsilon_0 \hbar^2 \kappa} \quad (128)$$

With these definitions, the radial equation takes the form -

$$\frac{d^2 u}{dr^2} = \left[1 - \frac{\rho_0}{\rho} + \frac{l(l+1)}{\rho^2} \right] u \quad (129)$$

We can find out the asymptotic form of the solutions to be -

$$u(\rho) \sim Ae^{-\rho}, \text{ for large } \rho$$

$$u(\rho) \sim B\rho^{l+1}, \text{ for small } \rho$$

Hence, we let,

$$u(\rho) = \rho^{l+1} e^{-\rho} v(\rho) \quad (130)$$

Putting this in the radial equation and simplifying, we get,

$$\rho \frac{d^2 v}{d\rho^2} + 2(l+1-\rho) \frac{dv}{d\rho} + [\rho_0 - 2(l+1)]v = 0 \quad (131)$$

We assume a power series solution for $v(\rho)$ as -

$$v(\rho) = \sum_{j=0}^{\infty} a_j \rho^j \quad (132)$$

It can be shown that the coefficients follow this recursion formula -

$$a_{j+1} = \left\{ \frac{2(j+l+1) - \rho_0}{(j+1)(j+2l+2)} \right\} a_j \quad (133)$$

To ensure that $u(\rho)$ does not blow up for large ρ , the series must terminate, i.e. there must exist some $j_{\max}$, so that $a_j = 0 \forall j > j_{\max}$.

Then we have from equation (133), that,

$$\rho_0 = 2(j_{\max} + l + 1)$$

Define

$$n = j_{\max} + l + 1 \quad (134)$$

so that

$$\rho_0 = 2n \quad (135)$$

Using equations (135) and (128), we get an expression for the allowed energies -

$$E_n = - \left[\frac{m}{2\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 \right] \frac{1}{n^2} = \frac{E_1}{n^2}, \quad n = 1, 2, 3, \dots \quad (136)$$

This is the famous **Bohr's formula**. The **ground state energy**, $E_1 = -13.6$ eV.

Let us define the **Bohr radius** a so that

$$\rho = \frac{r}{an} \quad (137)$$

We can find out the expression for a as -

$$a = \frac{4\pi\epsilon_0\hbar^2}{me^2} = 0.529 \times 10^{-10} \text{m} \quad (138)$$

Using equation (133) we can calculate the coefficients of the $v(\rho)$, upto a global constant a_0 , which can later be determined through normalization.

Hence we can find out $R(r)$. We can thus write the overall wave function $\Psi_{n,l,m}(r, \theta, \phi) = R_{n,l}(r)Y_{l,m}(\theta, \phi)$.

Let us explicitly compute the ground-state wave function ($n = 1$).

$$\begin{aligned} \rho &= \frac{r}{a} \\ R_{1,0}(r) &= \frac{1}{r} u(r) \\ &= \frac{1}{r} \rho e^{-\rho} a_0 \\ &= \frac{a_0}{a} e^{-r/a} \end{aligned}$$

Normalize it to deduce that $a_0 = 2/\sqrt{a}$. We know that $Y_{0,0} = 1/\sqrt{4\pi}$, so we get -

$$\Psi_{1,0,0}(r, \theta, \phi) = \frac{1}{\sqrt{\pi a^3}} e^{-r/a} \quad (139)$$

This wave function corresponds to the **1s orbital**.

Similarly, wave functions for the excited states can be found too, which correspond to higher orbitals.

6 Angular Momentum

The **angular momentum operator** in quantum mechanics is defined as -

$$\mathbf{L} = \mathbf{r} \times \mathbf{p} \quad (140)$$

which is to say,

$$L_x = yp_z - zp_y, \quad L_y = zp_x - xp_z, \quad L_z = xp_y - yp_x \quad (141)$$

6.1 Properties of Angular Momentum

- The angular momentum operators are Hermitian, i.e.,

$$L_x^\dagger = L_x, \quad L_y^\dagger = L_y, \quad L_z^\dagger = L_z$$

It means that Angular Momentum is an observable.

- Any two components of the angular momentum do not commute with each other. In fact, we have -

$$[L_i, L_j] = i\hbar\epsilon_{ijk}L_k \quad (142)$$

where ϵ_{ijk} is the Levi-civita symbol. We may thus write -

$$[L_x, L_y] = i\hbar L_z, \quad [L_y, L_z] = i\hbar L_x, \quad [L_z, L_x] = i\hbar L_y$$

- The squared norm of the total angular momentum, $L^2 = L_x^2 + L_y^2 + L_z^2$, *does* commute with individual components.

$$[L^2, L_x] = 0, \quad [L^2, L_y] = 0, \quad [L^2, L_z] = 0 \quad (143)$$

We can thus find simultaneous eigenstates of L^2 and L_z .

6.2 Eigenvalues and Eigenstates of Angular Momentum

Since L^2 and L_z commute, we can find simultaneous eigenstates of them. Define -

$$L_\pm = L_x \pm iL_y \quad (144)$$

Then it is easy to show that -

If f is an eigenstate of L^2 with eigenvalue λ , then $L_\pm f$ is also an eigenstate of L^2 with the same eigenvalue λ .

If f is an eigenstate of L_z with eigenvalue μ , then $L_\pm f$ is also an eigenstate of L_z with new eigenvalue $\mu \pm \hbar$.

Eigenvalues of L_z -

We can keep acting the L_+ operator on eigenstates to keep getting new eigenstates of L_z . However, in order to ensure that the z-component on angular momentum does not exceed the total, this process must terminate at some eigenstate f_t , i.e., there must exist eigenstate f_t so that $L_+ f_t = 0$.

We write the maximum eigenvalue of L_z , as $l\hbar$, where l is the **azimuthal quantum number**. For the same reason, the sequence of eigenvalues of L_z must also terminate from the bottom side, i.e., there exists an eigenstate f_b such that $L_- f_b = 0$.

The minimum eigenvalue of L_z is $-l\hbar$.

Hence the eigenvalues of L_z are located at steps of $\hbar$. So we label the eigenvalues as $m\hbar$, where m , called the **magnetic quantum number**, goes from $-l$ to l in **integer steps**. So for any given value of l , m can assume $2l + 1$ possible values. Moreover, if N is the number of steps taken to get from $-l$ to l , then we must have $-l + N = l$, which means -

$$l = \frac{N}{2} \quad (145)$$

Hence, we see that l is allowed to take integer values ($l = 0, 1, 2, \dots$) or half-integer values only ($l = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots$). However, we know from section (5.2) that l must only take integer values, so we drop the half-integer values.

The half-integer values have significance for **spin angular momentum**, which we shall discuss in the upcoming section about Spin.

Eigenvalues of L^2 -

It can be shown that the eigenvalues of L^2 are -

$$\lambda = l(l + 1)\hbar^2 \quad (146)$$

Eigenstates of L^2 and L_z -

It can be shown that the eigenstates of L^2 and L_z are the spherical harmonics. Clearly, the eigenstates are labelled by the quantum numbers l and m .

7 Spin

The angular momentum we discussed in the previous section is more precisely called the **orbital angular momentum**. It is associated to the motion of the particle around the origin (e.g., motion of electron around nucleus).

Quantum particles also carry an *intrinsic* angular momentum, called the **spin angular momentum**.

The quantum concept of 'spin' is not completely analogous to classical spin, since classical bodies with spin have constituent particles that are revolving about some axis. However, quantum particles like electrons do not have smaller constituent parts, and are assumed to be structureless point particles.

7.1 Algebra of Spin

The algebraic theory of spin is completely similar to that of orbital angular momentum.

- We **postulate** that the spin operators satisfy the following commutation relations -

$$[S_x, S_y] = i\hbar S_z, \quad [S_y, S_z] = i\hbar S_x, \quad [S_z, S_x] = i\hbar S_y \quad (147)$$

- S^2 commutes with S_z , i.e., $[S^2, S_z] = 0$.
- We can also define $S_{\pm} = S_x \pm iS_y$ operators, in the same way as we did for orbital angular momentum.
- The **eigenvalues** of S_z are of the form $m\hbar$, where $m \in \{-s, -s + 1, \dots, s - 1, s\}$. Here, s is called the **spin quantum number**.

- Unlike the case of orbital angular momentum, there is no reason to reject half-integer values of s . Hence s can take integer or half-integer values. For a given value of s , m can take $(2s + 1)$ values, ranging from $-s$ to s in steps of 1.
For example, for $s = 2$, $m \in \{-2, -1, 0, 1, 2\}$.
For $s = \frac{5}{2}$, $m \in \{\frac{-5}{2}, \frac{-3}{2}, \frac{-1}{2}, \frac{1}{2}, \frac{3}{2}, \frac{5}{2}\}$ and so on.
- The **eigenvalues** of S^2 are $s(s + 1)\hbar^2$.

It turns out that every elementary particle has a *fixed* value of s . Pi-mesons have spin 0, electrons have spin 1/2 and so on. This is unlike the case of orbital angular momentum, where an electron can have any integer value of the orbital angular momentum quantum number l .

7.2 Spin 1/2

Eigenvectors $|s, m\rangle$ of the spin operator are labeled by the quantum numbers s and m . For spin 1/2 particles, s is fixed at 1/2, and m can be either -1/2 or 1/2. Hence, for spin 1/2, there are only two eigenvectors, which are denoted by $|\frac{1}{2}, \frac{1}{2}\rangle$ (also called spin-up, and sometimes written as $\uparrow$) and $|\frac{1}{2}, \frac{-1}{2}\rangle$ (also called spin-down, sometimes written as $\downarrow$).

Hence, the vectors that the spin operators act on, lie in a 2-dimensional vector space. We can choose the eigenvectors of S_z to be the basis. Using this basis, we can write the general state of a spin 1/2 particle as a column vector (called a **spinor**)-

$$\chi = \begin{pmatrix} a \\ b \end{pmatrix} = a|\frac{1}{2}, \frac{1}{2}\rangle + b|\frac{1}{2}, \frac{-1}{2}\rangle$$

Since we are working with a 2-dimensional vector space, the spin operators can be represented as 2x2 matrices. It can be shown that the matrix representation of the spin operators is $\mathbf{S} = (\hbar/2)\boldsymbol{\sigma}$, where

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (148)$$

The $\boldsymbol{\sigma}$ matrices are called the **Pauli spin matrices**. These matrices are Hermitian, which must be true because spin is an observable.

The eigenspinors of S_z are $|\frac{1}{2}, \frac{1}{2}\rangle$ and $|\frac{1}{2}, \frac{-1}{2}\rangle$, which can be written as $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ respectively.

If we measure S_z on any general state $\begin{pmatrix} a \\ b \end{pmatrix}$, then the outcomes may only be $\hbar/2$ with probability $|a|^2$, and $-\hbar/2$ with probability $|b|^2$. Since these are the only probabilities, we must have -

$$|a|^2 + |b|^2 = 1 \quad (149)$$

i.e., the spinor must be normalized.